Airline Customer Analytics Dashboard Overview – Group 12

This dashboard delivers a comprehensive, filtered analysis of Canadian airline customer behavior, value, and demographics.

I. Key Performance Indicators (KPIs) and Summary Metrics

The top section aggregates four critical metrics based on the current filter selection, providing immediate insight into overall performance:

- **Median Flights**: Average number of flights per customer.
- **Median Distance (KM):** Average travel distance in kilometers.
- **% w/ Companions**: Percentage of flights booked with a companion.
- **Median Dollar Cost Points Remaining**: Average remaining loyalty point value (in dollars) available to customers.

II. Detailed Analysis Sections

Flight Metrics Over Time - This trend line chart visualizes four metrics:

- **Flights**: Total number of flights taken.
- **Flights w/ Companions**: Number of flights booked with companions
- **Distance (Thousands of KM)**: Total distance traveled
- **Dollar Cost (Points Remaining):** Monetary value of loyalty points remaining by customers

Categorical Overview - Provides segmentation insights across key groups: Marital status, enrollment type, loyalty status, gender, location, code.

Customer Value and Enrollment - Focuses on: Customer Lifetime Value Distribution, Income, Enrollment Date, Cancellation Date

Customer Distribution by City- Geographic mapping to visualize customer density and travel patterns across specific cities.

III. Applicable Filters

The dashboard is highly interactive, allowing users to drill down into specific segments using the following filters: **Years** (2019,2020,2021), **Provinces/States** (Nova Scotia, Ontario, Prince Edward Island...), **Cities** (Kelowna, Moncton, Peace River.....), **Enrollment Types**(2021 promotion, standard), **Marital Status**(divorced, married, single), **Loyalty Status**(Aurora, nova, star), **Education Levels** (Bachelor, college...).

To access the **dashboard** online**,** click this link: https://aiai-dashboard-dm.streamlit.app/

To access the **code** (see the delivery folder is the Dashboard.py) to run the dashboard local, go to command prompt and in the folder where the Dashboard.py is and write: **python -m streamlit run Dashboard.py**

Note: The data folder needs to have the csv files that are created running the EDA.ipynb notebook. And of course, it needs to have an environment with streamlit and plotly + the other libraries installed to work on the classes.